

Topological grid as an iterative selection criterion in segmentation of Herculaneum scrolls

A complementary tool proposal for the Vesuvius Challenge pipeline

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Abstract

This proposal presents a geometric validation tool to support the segmentation process of the Herculaneum scrolls. The central idea is that any classical writing on papyrus or parchment — whether Greek, Latin, Hebrew, Demotic or other ruled scribal traditions— produces, when the segmented surface is virtually unrolled, a regular spatial grid whose structural signature can be measured quantitatively. This signature —which we call the *topological grid*— constitutes a physical invariant independent of the specific textual reading, and therefore can be used to evaluate and rank automatically between alternative segmentations during a layer-by-layer growth process. Empirical validation on two versions of the Grand Prize 2023 (PHerc. Paris 4) confirms that the grid exists, is measurable, autocalibrates its parameters from the scroll itself, and discriminates between different versions produced by different ML architectures. The tool integrates as a ranking filter into any existing segmentation pipeline without requiring modifications to it.

1. Context

The Vesuvius Challenge team has identified segmentation —the tracing of the papyrus surface within the CT volume— as the economic bottleneck for scaling the reading of the entire Herculaneum corpus. Current methods (Henderson 2025, Thaumato Anakalyptor, Volume Cartographer, VC3D) approach the problem with purely geometric computer vision techniques: the surface is fitted to the volume using local information about density, continuity and smoothness, without incorporating knowledge about the textual content the surface bears.

In parallel, ink detection models (the three from Grand Prize 2023, subsequent iterations) work post hoc on already-segmented surfaces. The two phases —segmentation and ink detection— operate sequentially without feedback. This separation is reasonable as an initial simplification, but it leaves unexploited an information that the detected ink itself contains: *the spatial regularity inherent to writing*. Any correctly segmented and unrolled scroll must show a regular two-dimensional grid, because ancient scribes wrote that way. An incorrect segmentation breaks that regularity in detectable ways, even if the subsequent ink detection produces individually plausible outputs.

2. The topological grid as a measurable invariant

Classical writing on papyrus or parchment has a highly regular geometric structure, dictated by the practical conventions of scribes. This applies to the Greek of the Herculaneum scrolls, but also, with different parameters, to Latin, Hebrew, Demotic and other ruled writing traditions. The relevant regularities are three:

- The **text lines** are parallel, perpendicular to the scroll axis, and are spaced regularly. The specific period depends on the scroll and the scribe (in Greek philosophical scrolls from Herculaneum ~5 mm is observed, but the method does not assume this value; it extracts it from the data itself).
- The **letters** within each line are distributed with approximately constant spacing, forming a periodic sequence. In Herculaneum Greek the period is ~3 mm; in other traditions it will be different, but periodicity exists whenever the writing is ruled.

- The **columns** (in scrolls that have them) are rectangles parallel to the axis, separated by an intercolumnium of approximately constant width. The sum column+intercolumnium repeats along the scroll. Specific values vary between traditions: in Greek philosophical scrolls of Herculaneum periods of ~10–15 mm are observed; in other corpora they may differ but the periodicity persists.

On the 2D surface of the unrolled papyrus, this translates into a **two-dimensional grid** with three well-defined spatial periodicities. This grid is what we call the *topological grid*: a physical invariant that depends only on the scroll and the scribe, not on the specific textual content. A correct segmentation —whatever it may be— produces this grid. An incorrect segmentation breaks it in detectable ways.

The important point is that this structural signature can be measured quantitatively through spectral analysis. The three periodicities appear as peaks in the Fourier spectrum of the ink image on the unrolled surface. If the peaks are where they should be and sufficiently prominent, the grid is present. If not, it is broken.

3. Application: iterative ranking during layer-by-layer growth

The proposed operational application does not require replacing existing segmentation pipelines. It is inserted as a **ranking criterion between alternatives** during the natural growth process of segmentation from outside to inside the scroll:

1. The segmentation grows layer by layer, starting from the outer windings (where geometry is clear) towards the inner ones (where deformation and compression make the work more difficult).
2. At each step, the segmentation algorithm generates **several candidates** for the next layer: different radial boundaries, different treatments of discontinuities, different resolutions of local ambiguities.
3. Each candidate is subjected to the topological grid test: it is virtually unrolled and its local structural signature is measured.
4. The candidate that produces the most coherent grid is accepted as the segmentation of that layer, and the process advances to the next.
5. If none of the candidates exceeds a minimum coherence threshold, the zone is flagged for manual review. This replaces exhaustive visual inspection with a focused alert.

The advantage of this approach is operational: it does not require changing the segmentation architecture, only adding a selection criterion when the algorithm encounters several possibilities. It is an *automated arbiter between alternatives* produced by the existing pipeline, not a competitor to it.

4. Empirical validation

4.1. The grid exists and is measured globally

On the unrolled surface image of Grand Prize 2023 (TimeSformer version by Youssef Nader), Fourier analysis produces three spectral peaks at the positions predicted by the framework:

Quantity	Predicted by papyrology	Detected (autocalibrated)	Match
Line period (Y)	~5 mm	4.77 mm	✓
Letter period (X)	~3 mm	3.77 mm	✓
Column period (X)	~10–15 mm	11.30 mm	✓

Table 1. Spatial periods detected in Grand Prize 2023 (Paris 4). The three spectral peaks are located automatically without assuming fixed values: the method extracts the signature from the scroll itself. The values are consistent with the papyrology documented for Herculaneum philosophical scrolls.

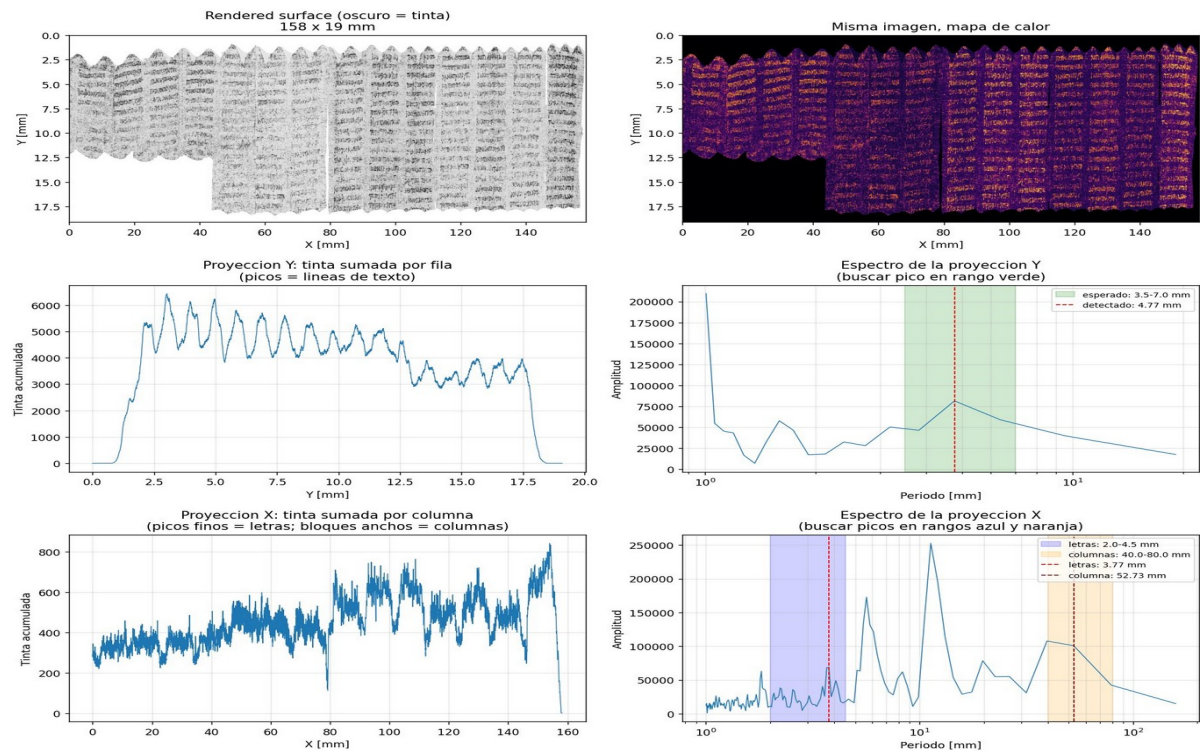


Figure 1. Fourier analysis on the unrolled surface of Paris 4. The three upper panels show the original image, the ink projection along Y (where the regular oscillation of lines is visible) and the corresponding spectrum with the expected peak at ~5 mm. The lower panels show the X projection and its spectrum, with clear peaks for letters and columns. Detection is automatic and does not assume predefined values.

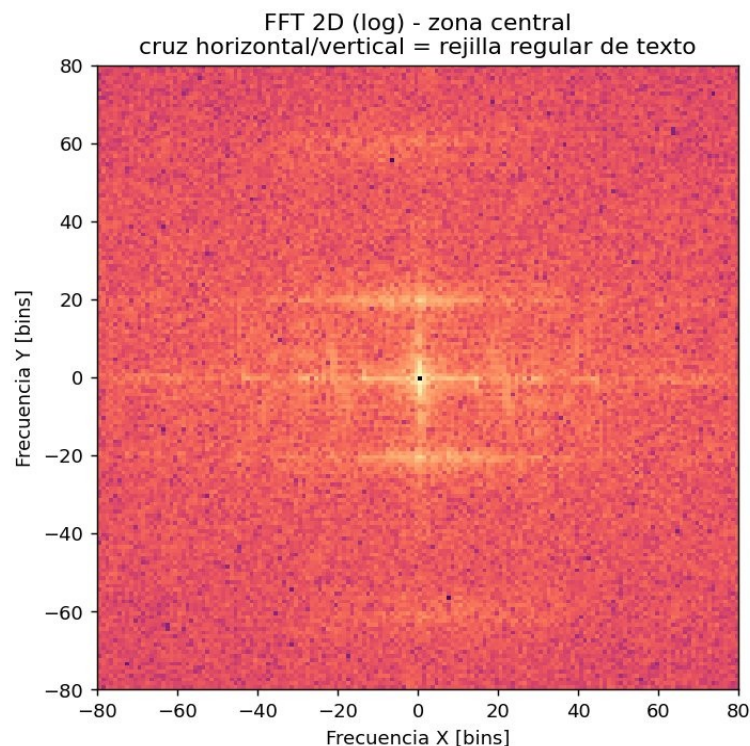


Figure 2. 2D FFT of the rendered surface of Paris 4 (logarithmic, central zone). The horizontal and vertical cross structure, with secondary harmonics visible at double frequencies, is the unequivocal signature of a regular two-dimensional grid. If the rendered surface were noise, the 2D spectrum would be an amorphous blob without orthogonal structure.

4.2. The signature is measured locally and discriminates between versions

The grid can be measured in local windows using a spectral prominence metric: in each region of the surface, how much each expected peak stands out above the local spectrum background is evaluated. Applying this metric to the two published versions of Grand Prize 2023 (TimeSformer and squeeze-net, produced by two different ML architectures on the same geometric segmentation), comparable topological maps are obtained.

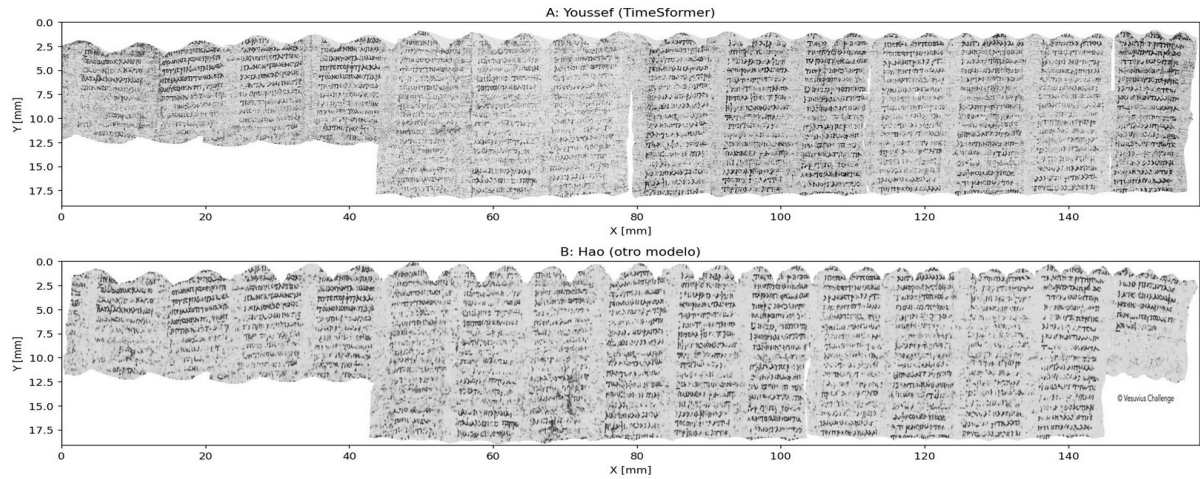


Figure 3. The two versions of the rendered surface of Paris 4 published by the Grand Prize 2023 team. The upper one (Youssef, TimeSformer model) and the lower one (Hao, another architecture) operate on the same geometric segmentation but produce different ink detections. They allow a direct test of the framework: if the topological grid is discriminative, it must produce different local maps.

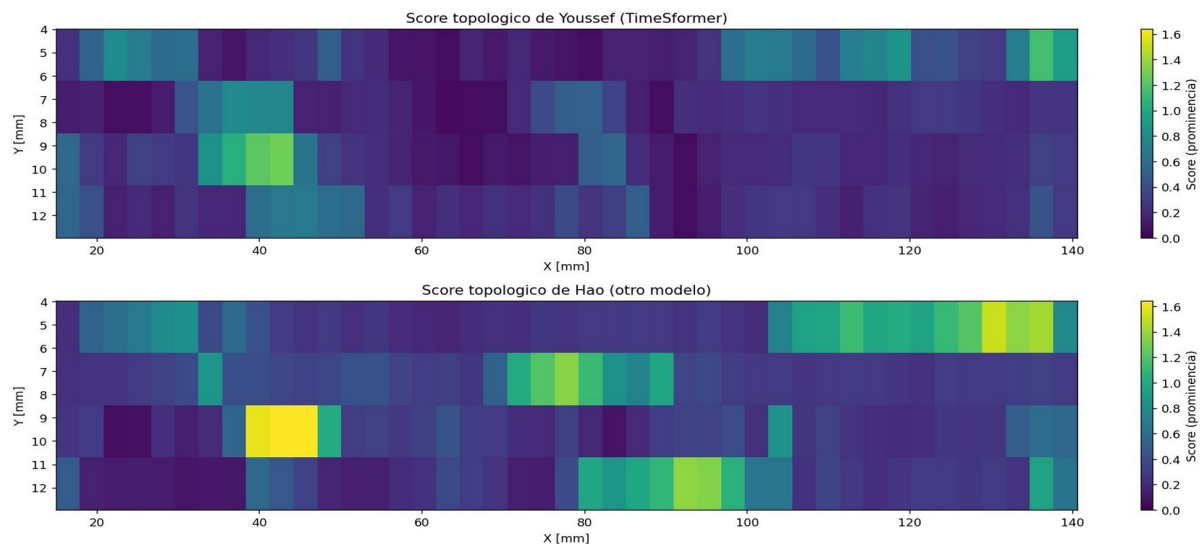


Figure 4. Local topological maps of the two versions, with the same color scale. The metric autocalibrates the common signature from both images and then measures local spectral prominence in each window. The two versions show different patterns of structural coherence: each ML model has zones where it detects clearer structure than the other.

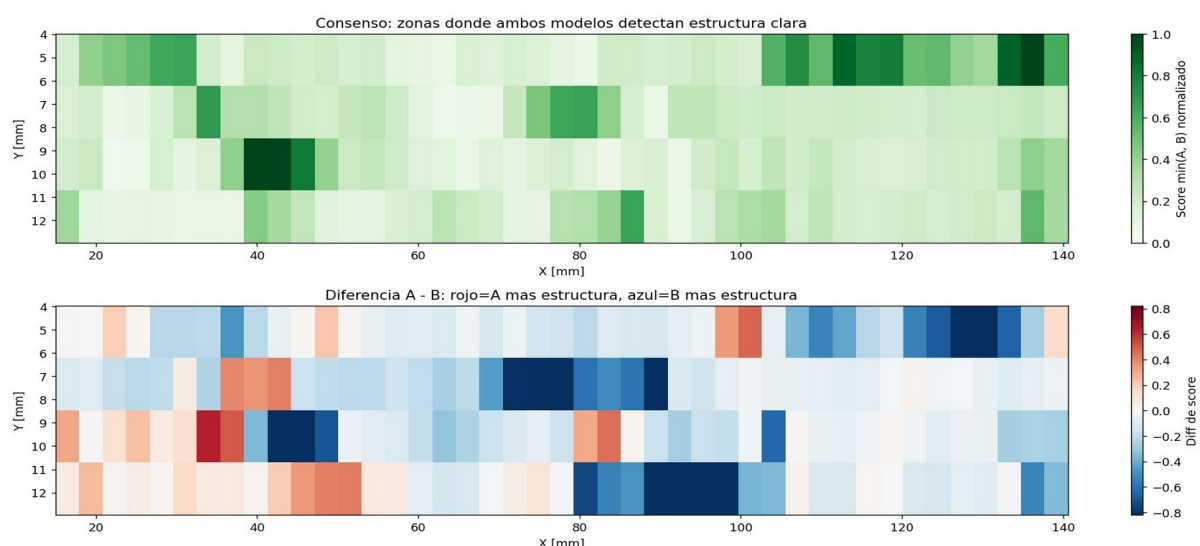


Figure 5. Upper panel: consensus zones between the two models (more intense green = both detect clear structure), natural candidates for robust readings. Lower panel: divergence (red where Youssef detects more, blue where Hao detects more). The zones of maximum divergence are those that would require papyrological review to determine which of the detections is real signal and which is artifact.

The result has two operational readings:

- **Consensus zones** between models represent readings where structure is robust. They are natural candidates to prioritize in the papyrological validation flow.
- **Divergence zones** are those that require attention: one of the models may be capturing real signal that the other misses, or one may be generating spurious structure. The topological grid detects and localizes these divergences automatically, which currently can only be done through visual inspection.

5. Implementation

The tool consists of a small self-contained Python module (numpy, scipy, matplotlib, pillow) that operates on any rendered surface in standard image format. Its operations are:

1. **Global calibration:** automatic detection of the three dominant periods of the scroll in question, without assuming predetermined values. The signature is learned from the scroll itself.
2. **Local mapping:** window sweep with spectral prominence calculation in each. Result: 2D topological map of coherence.
3. **Comparison between alternatives:** given two or more segmentations of the same scroll (or two or more ink detections on the same segmentation), it produces a differential map of consensus and divergence.
4. **Ranking:** an alternative with higher average score in a given region is preferred over another with lower score in the same region.

Computation is light: complete analysis of a rendered surface of Paris 4 size (158×19 mm at 8 $\mu\text{m}/\text{pixel}$ resolution) runs in less than a minute on a standard laptop. The tool can be incorporated into the segmentation pipeline as a callable subroutine when the main algorithm generates alternatives requiring arbitration.

6. Declared limitations

- **Calibration with papyrological ground truth.** The topological grid measures geometric regularity, not textual correctness. A segmentation may produce a coherent grid and still contain letter identification errors (coherent hallucination of the ML model). The correlation between topological score and final papyrological quality can only be validated with the internal data of the Vesuvius Challenge: the rendered surfaces already read by papyrologists would be the natural calibration set.
- **Per-scribe applicability.** The grid parameters (exact periods) vary between scribes and between scrolls. The tool autocalibrates these parameters for each scroll from the highest-quality zones, but requires a minimum initially well-segmented region for calibration. For scrolls completely compressed without an accessible clean zone, autocalibration does not work and an a priori value would be needed.
- **False positives in atypical zones.** Margins, intercolumnia, physical tears and illustrations legitimately lack the grid. The tool must be combined with a prior classification of zone type to avoid penalizing these regions; combining the topological score with total ink density resolves most cases.

7. Suggested next steps

If the proposal is of interest to the team, the natural next steps would be:

1. **Quantitative calibration** of the topological score against known papyrological reading rates, on already-validated rendered surfaces. This converts the score into a quantitative predictor and not just a qualitative one.
2. **Integration with a specific segmentation pipeline** (Thaumato Anakalyptor or whichever is preferred). Implement the ranking subroutine at the points of the algorithm where candidates are generated.
3. **Application to the Progress Prize "wish list"**: automatic detection of problematic zones in current rendered surfaces could reduce the manual review cost and improve the effective 90% coverage.

On my side I am available to collaborate on any of these steps, share the code of the tool (Python, MIT/Apache), or discuss variants. The tool is documented and can be run on any public rendered surface without additional dependencies.

The topological grid will not read new letters that the ML does not detect, nor will it replace existing segmentation. What it offers is a quantitative, automatable and reproducible metric to arbitrate between alternatives during the process, focus human review where it is really needed, and provide an objective measure of robustness for the readings the team already produces. It is a small but, I believe, genuinely useful piece for scaling the work to the remaining scrolls.